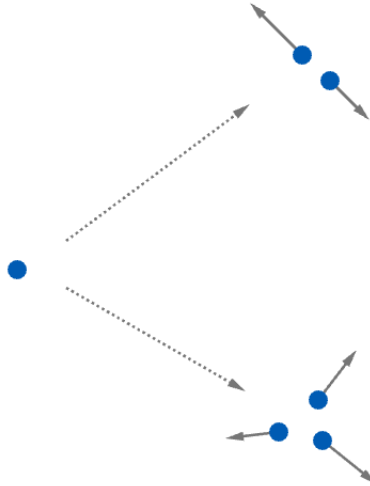


2008 F=ma Exam: Problem 21

Kevin S. Huang



- (a) True in both cases by the conservation of linear momentum. The velocity vectors of the daughter particles must lie in a single line in the two-body case but that is a special case of being in a single plane.
- (b) True in the two-body case but false in the three-body case. We have conservation laws

$$K = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$

$$0 = m_1v_1 + m_2v_2$$

for the two-body case which is 2 equations for 2 unknowns. The three-body case has more unknowns than equations.

- (c) False in both cases since the masses are unknown.
- (d) True in both cases by the conservation of linear momentum.
- (e) Not applicable.

Thus, the answer is B.